

Robust Near-field Beamforming based on Probability Constraints and Aperture Perturbations

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ABSTRACT

In this paper, we consider robust near-field beamforming for the uplink of SIMO mmWave networks under aperture perturbations of the antenna array at the base station (BS) with unknown interference. Note that the worst-case approach can be overly pessimistic because the probability of the actual worst-case errors occurring in practice may be extremely low. Therefore, we adopt a more flexible and efficient approach for the implementation of robust near-field beamforming using probabilistic constraints. We cast the probabilistic-constrained optimization problem as a convex second-order cone program (SOCP). The proposed beamformer has the advantage of being less conservative. Simulation results confirm that the proposed robust near-field beamforming algorithm outperforms the available benchmarks in the literature.

Index Terms— robust beamforming, aperture perturbation, near-field beamforming, millimeter frequency range

1. INTRODUCTION

The deployment of an extremely large number of antennas at the base stations (BSs) and the exploitation of higher frequencies has been recognized as a promising solution for future wireless systems [1, 2]. Due to the propagation conditions at these high frequencies and large antenna arrays at the BS, the near-field region of the BS is expanded. Therefore, this will cause the communicating devices to operate within the BS antenna's near-field [3, 4]. As a result, the far-field assumption used regularly in conventional wireless systems does not hold. Over the recent years, there has been an increased interest within the research community for research on near-field beamforming. Radiative near-field propagation takes place between the Fraunhofer distance and the Fresnel distance of large-scale antenna arrays operating at mmWave frequencies [5, 6]. The analyses in [3, 7] show that the near-field distance can be several dozens of meters for relatively small antenna arrays operating at mmWave and terahertz (THz) frequencies.

At the same time, due to the potentially large size of antenna arrays operating at mmWave frequencies, there might be different reasons for the emergence of perturbations in the aperture geometry. For example, these errors might occur during the installation of the antenna array or due to the impact of atmospheric effects on the antenna array such as high wind. Additionally, due to the size of the antenna array, it is difficult to ensure a perfect aperture geometry during the installation of the antenna, which might consist of multiple panels being stacked together. As a result, this leads to inevitable aperture deformations that degrade the overall performance of the communication system. For instance, the authors in [8, 9] have shown that the change of curvature influences the antenna directivity and the level of the sidelobes.

A review of the state-of-the-art has shown that most of the conventional methods for robust beamforming are designed for far-field scenarios. Moreover, most of them are focused on the robustness against steering vector mismatch, which is caused by an estimation error of the angle of arrival (AoA) or angle of departure (AoD) [10, 11, 12]. For example, the authors in [10] proposed a robust algorithm based on the prior information about the bound on the steering vector mismatch to avoid self-nulling of the desired signal even in the case of worst-case perturbations. On the other hand, the authors in [13] considered a beamforming design robust to aperture deformations caused by thermal deformations. Unfortunately, this work has a limited application for the problem we consider, since the system model is based on the far-field assumption and considers non-terrestrial networks. There are also a few references related to the design of robust beamforming for near-field users. The authors in [14] proposed a near-field beamforming algorithm robust to steering vector mismatches, which is a similar problem as in [10]. In [15], the authors proposed a near-field beamforming algorithm with robustness against distance errors. In [16], the authors considered a near-field beamforming design that is robust against aperture perturbations. However, the design is based on a worst-case performance optimization. This is impractical as the worst-case approach is overly conservative since the probability of the actual worst-case errors occurring in practice may be extremely low. Motivated by this fact, we consider a different approach in this paper to obtain another type of robust formulation using more flexible and realistic probabilistic constraints, based on a statistical steering vector error model.

In this paper, we study a probabilistic-constrained robust near-field beamforming for the uplink of SIMO mmWave networks under aperture perturbations of the BS antenna array. The resulting probabilistic-constrained optimization problem is cast as a convex second-order cone program (SOCP) that can be easily solved using available off-the-shelf tools.

2. SYSTEM MODEL

We consider the uplink of a mmWave communication system where the BS is equipped with a uniform rectangular array (URA) in the x - y plane consisting of M_x and M_y antennas along the x - and y -directions, respectively, such that $M = M_x \cdot M_y$ is the total number of BS antennas. We assume that the phase center of the BS antenna array is at the origin of the coordinate system as shown in Fig. 1. The inter-element spacing between the elements along the x -axis and the y -axis is denoted as Δ_x and Δ_y , respectively.

A single antenna user equipment (UE) communicates with the BS in the radiative near-field of the BS via a line-of-sight (LoS) connection. The spherical coordinates of the UE are given as $(\theta_{rx}, \phi_{rx}, r_{rx})$, where θ_{rx} and ϕ_{rx} are the azimuth and elevation angles in the direction of the phase center of the transmitter, while r_{rx} is the

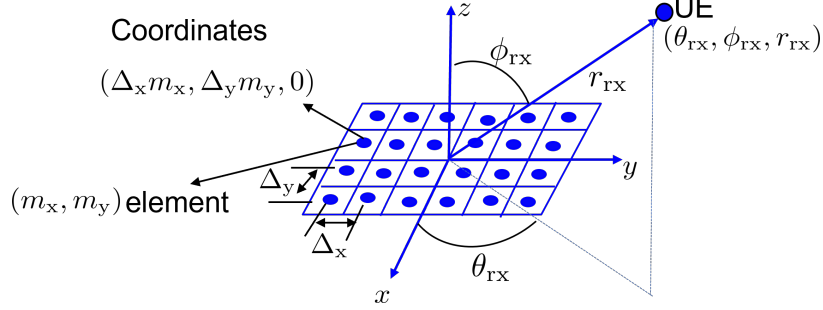


Fig. 1: Configuration and orientation of the URA at the BS

distance towards it. The output of the beamformer is given as

$$y(k) = \mathbf{w}^H \mathbf{x}(k), \quad (1)$$

where $\mathbf{w} \in \mathbb{C}^{M \times 1}$ is denotes the vector of beamforming weights, $\mathbf{x}(k) \in \mathbb{C}^{M \times 1}$ is the input vector, and k denotes the time index. The input vector can be described as

$$\mathbf{x}(k) = \mathbf{a}s(k) + \mathbf{i}(k) + \mathbf{n}(k) \quad (2)$$

where $\mathbf{a} = \text{vec}\{\mathbf{A}\} \in \mathbb{C}^{M \times 1}$ is the vectorization of the near-field steering matrix $\mathbf{A} \in \mathbb{C}^{M_x \times M_y}$ comprising the LoS connection between the UE and every element of the BS antenna array. Furthermore, $s(k)$ is the desired signal, $\mathbf{i}(k)$ is the interference, and $\mathbf{n}(k) \in \mathbb{C}^{M \times 1}$ is the noise assumed as zero-mean circularly symmetric complex Gaussian.

The (m_x, m_y) element of the near-field steering matrix $\mathbf{A}$ where $0 \leq m_x \leq M_x - 1$, $0 \leq m_y \leq M_y - 1$ can be described as

$$\mathbf{A}(m_x, m_y) = \frac{1}{\delta_{m_x, m_y}} e^{-j \frac{2\pi}{\lambda} \delta_{m_x, m_y}} \in \mathbb{C}, \quad (3)$$

where δ_{m_x, m_y} is the distance between the (m_x, m_y) element of the BS antenna array and the receive antenna given as

$$\delta_{m_x, m_y} = \sqrt{(x_{\text{rx}} - m_x \Delta_x)^2 + (y_{\text{rx}} - m_y \Delta_y)^2 + (z_{\text{rx}} - 0)^2} \quad (4)$$

where $(x_{\text{rx}}, y_{\text{rx}}, z_{\text{rx}})$ represent the cartesian coordinates of the UE, which are respectively given as $x_{\text{rx}} = r_{\text{tx}} \cos \phi_{\text{tx}} \sin \theta_{\text{tx}}$, $y_{\text{rx}} = r_{\text{tx}} \sin \phi_{\text{tx}} \sin \theta_{\text{tx}}$, and $z_{\text{rx}} = r_{\text{tx}} \cos \theta_{\text{tx}}$. Using a first order approximation, (4) can be further simplified as

$$\delta_{m_x, m_y} = L_{\text{tx}} \sqrt{1 - \frac{2m_x \Delta_x x_{\text{rx}} + 2m_y \Delta_y y_{\text{rx}} - m_x^2 \Delta_x^2 - m_y^2 \Delta_y^2}{L_{\text{tx}}^2}} \quad (5a)$$

$$\approx L_{\text{tx}} - \frac{2m_x \Delta_x x_{\text{rx}} - m_x^2 \Delta_x^2}{2L_{\text{tx}}} - \frac{2m_y \Delta_y y_{\text{rx}} - m_y^2 \Delta_y^2}{2L_{\text{tx}}}, \quad (5b)$$

where $L_{\text{tx}} = \sqrt{x_{\text{rx}}^2 + y_{\text{rx}}^2 + z_{\text{rx}}^2}$ and we used the approximation $\sqrt{1+x} \approx 1 + \frac{x}{2}$ for small x .

3. PROBLEM FORMULATION

The signal-to-interference-plus-noise ratio (SINR) at the receiver can be written as [17]

$$\text{SINR} = \frac{\sigma_s^2 |\mathbf{w}^H \mathbf{a}|^2}{\mathbf{w}^H \mathbf{R}_{i+n} \mathbf{w}}, \quad (6)$$

where $\mathbf{R}_{i+n} = \mathbb{E}\{[\mathbf{i}(k) + \mathbf{n}(k)][\mathbf{i}(k) + \mathbf{n}(k)]^H\} \in \mathbb{C}^{M \times M}$ is the interference-plus-noise covariance matrix, σ_s^2 is the signal power, and $\mathbb{E}\{\cdot\}$ denotes the statistical expectation. The optimal weight vector can be found via a maximization of the SINR in (6) which is equivalently written as

$$\min_{\mathbf{w}} \quad \mathbf{w}^H \mathbf{R}_{i+n} \mathbf{w} \quad \text{s. t.} \quad \mathbf{w}^H \mathbf{a} = 1. \quad (7)$$

However, the matrix $\mathbf{R}_{i+n}$ is commonly replaced by the sample covariance matrix due to the unavailability of the true interference-plus-noise covariance matrix in practice. The sample covariance matrix given as

$$\hat{\mathbf{R}} = \frac{1}{N_s} \sum_{k=1}^{N_s} \mathbf{x}(k) \mathbf{x}(k)^H, \quad (8)$$

where N_s is the number of snapshots. Therefore, problem (7) can be rewritten as

$$\min_{\mathbf{w}} \quad \mathbf{w}^H \hat{\mathbf{R}} \mathbf{w}, \quad \text{s. t.} \quad \mathbf{w}^H \mathbf{a} = 1. \quad (9)$$

A direct solution to the minimum variance distortionless response (MVDR) problem in (9) is known as the sample matrix inversion (SMI) beamformer, where the optimal weight vector is given as

$$\mathbf{w}_{\text{SMI}} = \alpha \hat{\mathbf{R}}^{-1} \mathbf{a}, \quad (10)$$

and $\alpha = 1/\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{a}$. However, this solution essentially lacks the necessary robustness to handle a mismatch between the presumed and actual steering vectors. As a result, this leads to self-nulling of the desired signal. We denote the actual steering vector comprising the effect of the random perturbations of the surface on the BS antenna array as $\tilde{\mathbf{a}}$. The relationship between the actual and the presumed steering vector is given as [16]

$$\tilde{\mathbf{a}} = \mathbf{a} + \mathbf{e}, \quad (11)$$

where $\mathbf{a} = \text{vec}(\mathbf{A})$ and the vector $\mathbf{e} \in \mathbb{C}^{M \times 1}$ models the effect of the random perturbations on the steering vector due to aperture perturbations. The norm of the error vector is bounded by γ , i.e., $\|\mathbf{e}\| \leq \gamma$, which represents the worst-case bound of the perturbations. In [16], this bound was analytically derived as a function of the known bounds of the coordinate displacement. The error vector $\mathbf{e}$ is represented according to the aperture perturbation model in [16], where the m th element of $\mathbf{e}$ is given as $e(m) = \text{vec}(\Delta \mathbf{A}(m_x, m_y))$, where

$$\Delta \mathbf{A}(m_x, m_y) = \mathbf{p}_{m_x, m_y}^T \mathbf{q}_{m_x, m_y} \mathbf{A}(m_x, m_y), \quad (12)$$

$$\mathbf{q}_{m_x, m_y} = j \frac{2\pi}{\lambda} \left[\frac{(x_{\text{rx}} - m_x \Delta_x)}{L_{\text{rx}}}, \frac{(y_{\text{rx}} - m_y \Delta_y)}{L_{\text{rx}}}, \frac{z_{\text{rx}}}{L_{\text{rx}}} \right]^\top,$$

and $\mathbf{p}_{m_x, m_y} = [p_{m_x, m_y}^{\text{err}, x}, p_{m_x, m_y}^{\text{err}, y}, p_{m_x, m_y}^{\text{err}, z}]^\top$. The variable, $p_{m_x, m_y}^{\text{err}, i}$ is the random displacement along the i axis, $i \in \{x, y, z\}$ following a Gaussian distribution with zero mean and variance σ_i^2 . We denote $\mathcal{A}$ as the set of all possible realizations of the steering vector impacted by the random perturbations

$$\mathcal{A} = \{\tilde{\mathbf{a}} | \tilde{\mathbf{a}} = \mathbf{a} + \mathbf{e}, \|\mathbf{e}\| \leq \gamma\}. \quad (13)$$

As a result, the MVDR problem in (9) for the worst-case perturbation in the case of steering vector mismatch is written as

$$\min_{\mathbf{w}} \mathbf{w}^H \hat{\mathbf{R}} \mathbf{w}, \quad \text{s. t.} \quad \min_{\tilde{\mathbf{a}} \in \mathcal{A}} |\mathbf{w}^H \tilde{\mathbf{a}}| \geq 1. \quad (14)$$

The constraint in problem (14) will be satisfied by a worst-case vector $\tilde{\mathbf{a}}$ that lies on the boundary of the uncertainty set $\mathcal{A}$. A similar worst-case robust beamforming approach has been studied for near-field communication in [16, 14] and [10] for far-field communication. However, the worst-case approach can be overly pessimistic, since the probability that the actual worst-case errors occur in practice may be extremely low.

3.1. Robust Beamforming with Probability Specification

In this section, we adopt a more flexible and efficient approach for the implementation of robust near-field beamforming using probabilistic constraints. We approximate the bounded random uncertainty with a Gaussian model to obtain a tractable and less conservative robust design. We aim to ensure that the constraint is satisfied with a specified probability ρ , $\rho \in [0, 1]$. Here, we model the error vector $\mathbf{e}$ as an unknown random vector with a known statistical distribution similar to [18]. Therefore, the probabilistic-constrained minimum variance beamforming problem can be written as

$$\min_{\mathbf{w}} \mathbf{w}^H \hat{\mathbf{R}} \mathbf{w}, \quad \text{s. t.} \quad \Pr\{|\mathbf{w}^H \tilde{\mathbf{a}}| \geq 1\} \geq \rho, \quad (15)$$

where $\Pr\{\cdot\}$ is the probability operator and ρ is a quality of service (QoS) based preselected tolerable probability value. This value is related to the beamformer outage probability as $\rho = 1 - \rho_{\text{out}}$. Following [18, 19], we assume that the error vector $\mathbf{e}$ is drawn from a complex circularly symmetric Gaussian distribution with zero mean and covariance matrix $\mathbf{C}_e$, i.e.,

$$\mathbf{e} \sim \mathcal{CN}(\mathbf{0}_M, \mathbf{C}_e). \quad (16)$$

where the covariance matrix captures the second-order statistics of the uncertainties in the steering vector. Adopting a Gaussian assumption for the errors is realistic, since these errors are due to multiple independent perturbations. According to the Central Limit Theorem, the sum of multiple independent errors follows a Gaussian distribution. Due to the assumption about the independent perturbations in different coordinates, the covariance matrix is given as $\mathbf{C}_e = \text{diag}(\{\sigma_{e_m}^2\}_{m=1}^M)$, where $\sigma_{e_m}^2$ can be found as

$$\sigma_{e_m}^2 = \frac{4\pi^2}{(\delta_{m_x, m_y} \lambda)^2} \left(\frac{(x_{\text{rx}} - m_x \Delta_x)^2}{L_{\text{rx}}^2} \sigma_x^2 + \frac{(y_{\text{rx}} - m_y \Delta_y)^2}{L_{\text{rx}}^2} \sigma_y^2 + \frac{z_{\text{rx}}^2}{L_{\text{rx}}^2} \sigma_z^2 \right),$$

where $m = m_x + m_y$ and σ_x^2 , σ_y^2 , and σ_z^2 are the variances of the random displacement along the x-axis, y-axis, and the z-axis, respectively. As in [10], we assume that the steering vector errors

are reasonably small. Then from the triangle inequality [20], we have $|\mathbf{w}^H(\mathbf{a} + \mathbf{e})| \geq |\mathbf{w}^H \mathbf{a}| - |\mathbf{w}^H \mathbf{e}|$. Therefore, we can write the constraint in (15) as $\Pr\{|\mathbf{w}^H \mathbf{e}| \leq |\mathbf{w}^H \mathbf{a}| - 1\} \geq \rho$. As a result, the problem (15) is equivalently rewritten as

$$\min_{\mathbf{w}} \mathbf{w}^H \hat{\mathbf{R}} \mathbf{w}, \quad \text{s. t.} \quad \Pr\{|\mathbf{w}^H \mathbf{e}| \leq |\mathbf{w}^H \mathbf{a}| - 1\} \geq \rho. \quad (17)$$

Considering (16), the distribution of the random variable $\mathbf{w}^H(\mathbf{a} + \mathbf{e})$ can be written as

$$\mathbf{w}^H(\mathbf{a} + \mathbf{e}) \sim \mathcal{CN}\left(|\mathbf{w}^H \mathbf{a}|, \|\mathbf{C}_e^{1/2} \mathbf{w}\|^2\right). \quad (18)$$

Since the random variable $\mathbf{w}^H \mathbf{e}$ has a zero-mean circularly symmetric complex Gaussian distribution, the real and imaginary parts are real independent identically distributed (i.i.d.) Gaussian random variables, i.e.,

$$\text{Re}\{\mathbf{w}^H \mathbf{e}\} \sim \mathcal{CN}\left(0, \|\mathbf{C}_e^{1/2} \mathbf{w}\|^2 / 2\right), \quad (19)$$

$$\text{Im}\{\mathbf{w}^H \mathbf{e}\} \sim \mathcal{CN}\left(0, \|\mathbf{C}_e^{1/2} \mathbf{w}\|^2 / 2\right). \quad (20)$$

Next, we exploit the fact that the cumulative density function (CDF) of a Rayleigh-distributed random variable $z = \sqrt{u^2 + v^2}$ is given as $F(z) = 1 - e^{-z^2/2\sigma^2}$, where u and v are two real zero-mean Gaussian random variables with variance σ^2 . Then, following [18] the constraint in (17) can be recast as

$$\Pr\{|\mathbf{w}^H \mathbf{e}| \leq |\mathbf{w}^H \mathbf{a}| - 1\} = 1 - \exp\left(-\frac{(|\mathbf{w}^H \mathbf{a}| - 1)^2}{\|\mathbf{C}_e^{1/2} \mathbf{w}\|^2}\right) \geq \rho. \quad (21)$$

Furthermore, the inequality on the right-hand-side of (21) can be rewritten as

$$\|\mathbf{C}_e^{1/2} \mathbf{w}\| \leq \frac{1}{\sqrt{-\ln(1 - \rho)}} (|\mathbf{w}^H \mathbf{a}| - 1), \quad (22)$$

to satisfy the distortionless response constraint for the presumed steering vector (i.e., $|\mathbf{w}^H \mathbf{a}| \geq 1$). Since the cost function of (17) is unchanged when $\mathbf{w}$ undergoes an arbitrary phase rotation [10], we can select $\mathbf{w}$ such that $\text{Re}\{\mathbf{w}^H \mathbf{a}\} \geq 0$ and $\text{Im}\{\mathbf{w}^H \mathbf{a}\} = 0$. Therefore, the robust minimum variance beamforming problem with the probabilistic constraint in (15) can be rewritten as

$$\min_{\mathbf{w}} \mathbf{w}^H \hat{\mathbf{R}} \mathbf{w} \quad (23a)$$

$$\text{s. t.} \quad \|\mathbf{C}_e^{1/2} \mathbf{w}\| \leq \frac{1}{\sqrt{-\ln(1 - \rho)}} (|\mathbf{w}^H \mathbf{a}| - 1). \quad (23b)$$

The authors in [21] show that the optimization problem in (23) belongs to the class of SOCPs. This implies that the probabilistic-constrained beamforming problem can be approximated by a convex SOCP, which can be efficiently solved using interior point methods. **Complexity analysis:** Since the complexity per iteration of SOCP with n variables is $\mathcal{O}(n^3)$, the complexity of computing the covariance matrix $\hat{\mathbf{R}}$ is $\mathcal{O}(N_s M^2)$, and $\mathcal{O}(M)$ for computing the norm constraints, then the complexity of computing the proposed robust beamforming is $\mathcal{O}(N_s M^2 + M^{3.5} \log(1/\epsilon))$. Moreover, the complexity of the SMI scheme is $\mathcal{O}(M^3)$, and the complexity of SDP-based worst-case robust beamforming is $\mathcal{O}(M^6)$.

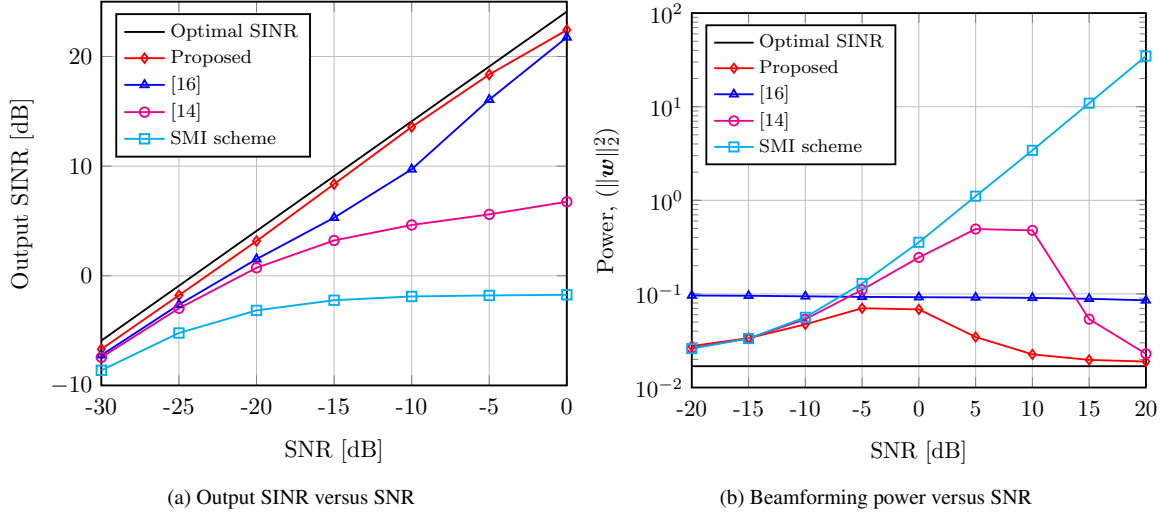


Fig. 2: Simulation results with $M_x = 64$ and $M_y = 4$ antennas and $N_s = 1000$.

4. NUMERICAL RESULTS

In this section, we show simulation results to evaluate the performance of the proposed and existing algorithms. We assume that the BS uses a URA with $M_x = 64$ and $M_y = 4$ antennas, which are uniformly spaced. For simplicity we assume that $\Delta_x = \Delta_y = \Delta = \lambda/2$ at a carrier frequency of 28 GHz. The azimuth and elevation angles of the desired UE are given as $\{90^\circ, 45^\circ\}$. We assume that there are two interferers whose azimuth and elevation angles are $\{50^\circ, 120^\circ\}$ and $\{120^\circ, 90^\circ\}$, respectively. The interference-to-noise ratio (INR) is 20 dB. The preselected probability value is given as $\rho = 0.95$, and the number of samples is $N_s = 1000$. We choose the same distance for the UEs in the radiative near-field of the BS antenna.

In our simulations, the perturbations are generated according to the Gaussian model in (16), where the standard deviation of the random displacement for each of the axes is given as $\{\sigma_x, \sigma_y, \sigma_z\} = \{0.01\lambda, 0.01\lambda, 0.05\lambda\}$. The proposed algorithm is compared with the optimal solution (optimal SINR) given as $\text{SINR}_{\text{opt}} = \sigma_s^2 \tilde{a}^H \mathbf{R}_{i+n} \tilde{a}$ according to [22], the sample matrix inversion (SMI) beamforming scheme in (10), the worst-case near-field beamforming with aperture perturbations given in [16], and the worst-case near-field beamforming design with a mismatch in the steering vector due to AoA errors according to [14]. For the steering vector mismatch algorithm, we modified the known bound given in [14], to take into account the number of sensors, $\epsilon = c\sqrt{M}$, $c = 0.01$.

Fig. 2a shows the SINR performance over the SNR for $N_s = 1000$ samples. The results are averaged over 1000 realizations. From the figure, it is observed that the proposed probabilistic-constrained robust near-field beamforming has a better performance compared to the existing schemes. The performance of the proposed beamformer demonstrates the important advantage of the proposed approach, which is less conservative than the worst-case beamforming in [16]. This is due to the fact that the proposed scheme explicitly specifies the parameters of the uncertainty region in terms of the covariance matrix and the outage probability. As a result, the proposed scheme has a lower power requirement in comparison with the other schemes, as shown in Fig. 2b. In Fig. 3, we depict the output SINR as a function of the number of snapshots. The considered SNR is

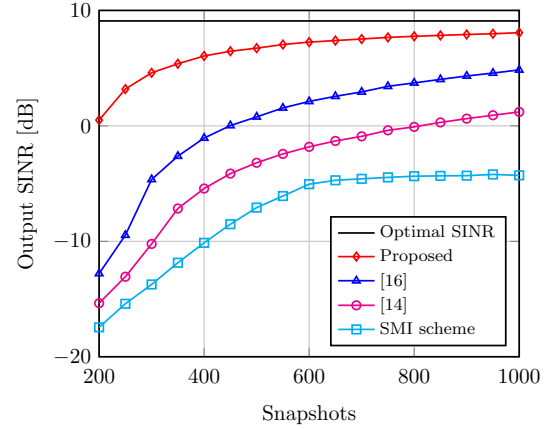


Fig. 3: Output SINR versus N_s , SNR = -15 dB

-15 dB, and the results are averaged over 1000 realizations. Here, the comparison of the convergence of the proposed robust near-field beamforming algorithm and the existing schemes is demonstrated. The proposed algorithm outperforms the existing solutions.

5. CONCLUSIONS

In this paper, we have proposed a probabilistic-constrained robust near-field beamforming algorithm for the uplink of a SIMO wireless communication system with unknown interference. The proposed algorithm demonstrates robustness to the steering vector error due to BS antenna array aperture perturbations. To obtain a solution, the proposed probabilistic-constrained beamforming problem is approximated by a convex SOCP. The proposed beamformer has the advantage of being less conservative than the worst case approach. It enables a prudent selection of the robustness parameters as compared with the worst-case approach by explicitly specifying them in terms of the covariance matrix and the outage probability. The simulation results confirm that the proposed robust near-field algorithm outperforms the benchmark schemes.

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